



List Comprehensions

Topics:

- Understanding List Comprehensions in Python
- Examples

List Comprehensions

[expression for element **in list]**

Applies the expression to each element in the list

You can have 0 or more for or if statements

If the expression evaluates to a tuple it must be in parenthesis

Examples

```
# program to multiply each element  
with 2
```

```
x = [10,20,30,40,50]
```

```
x1 = [i*2 for i in x]
```

```
print(x1)
```

Result:

```
[20, 40, 60, 80, 100]
```

Examples

```
# program to find the square of  
only even numbers.
```

```
x = [1,5,8,4,10,12,15,18,24]  
x1 = [i**2 for i in x if i%2==0 ]  
print(x1)
```

Result:

```
[64, 16, 100, 144, 324, 576]
```

Examples

```
# program to find the squares of even  
numbers and cube of odd numbers
```

```
x = [1,5,8,4,10,12,15,18,24]  
x1 = [i**2 if i%2==0 else i**3 for i in x ]  
print(x1)
```

Result:

```
[1, 125, 64, 16, 100, 144, 3375, 324, 576]
```

Examples

```
# program to find the squares of even  
numbers and cube of odd numbers
```

```
x = [1,5,8,4,10,12,15,18,24]  
x1 = [i**2 if i%2==0 else i**3 for i in x ]  
print(x1)
```

Result:

```
[1, 125, 64, 16, 100, 144, 3375, 324, 576]
```

Examples

```
# program with nested for statement
```

```
x = [[1,2,3,4],[5,6,7,8],[9,10,11,12]]
```

```
[j for i in x for j in i]
```

```
result:
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
```